

Pipelines Without Pipelines

How we built a data-pipeline engine with **no pipeline object**

A finite state machine you never wrote, implemented by the scripts themselves

First, the thing we're contrasting against

Windmill = run scripts (TS/Python/Go/SQL/...) as jobs.

Flows = wire scripts into a workflow.

A flow is a **declarative DAG**, fully defined upfront:

```
FlowValue {  
  modules: [ FlowModule, FlowModule, ... ]  // the whole graph, as data  
}
```

Branches, loops, retries, suspends, sub-flows — all just **data** in that tree.

`backend/windmill-types/src/flows.rs`

Flows are a finite state machine

Two structures, mirror images:

Definition (static)	State (dynamic)
<code>FlowValue.modules[]</code>	<code>FlowStatus.modules[]</code>
the graph	a cursor + per-step status
<code>flows.rs</code>	<code>flow_status.rs</code>

```
enum FlowStatusModule {  
    WaitingForPriorSteps, WaitingForExecutor,  
    InProgress, Success, Failure, WaitingForEvents,  
}
```

`FlowStatus { step: i32, modules: [...] }` → **persisted as JSONB after every step.**

The question

The engine is **generic**. The graph lives in the **definition**.

So what if there were **no definition**?

What if the graph were an **emergent property** of the scripts —
never written down anywhere?

Data pipelines: there is no pipeline

No `PipelineValue`. No DAG file. No orchestrator.

Just scripts that declare what they **touch**:

```
-- producer.sql
ATTACH 'datatable://main' AS db;
CREATE TABLE db.orders AS
  SELECT ... ;

-- consumer.sql
ATTACH 'datatable://main' AS db;
-- on datatable://main/orders
CREATE TABLE db.report AS
  SELECT ... FROM db.orders;
```

`producer` **writes** `orders`. `consumer` **reads** `orders`.

That edge is never declared. It's **parsed out of the SQL**.

The graph emerges from reads & writes

At deploy, we parse each script and write rows:

```
asset(path, kind, access_type, usage_path)
orders    datatable    W    f/producer
orders    datatable    R    f/consumer
```

The graph endpoint just **SELECT**s and joins these back together.

No compile step. Rename a script → graph updates instantly.

```
[trigger] → producer -writes-▶ (orders) -reads-▶ consumer -▶ (report)
```

Execution: dispatch, not orchestration

A flow has a conductor stepping `i → i+1`.

A pipeline has **none**. It's a chain reaction:

```
// after ANY top-level script succeeds:
dispatch_asset_triggers(job)
  → for each asset the script WROTE
    → find scripts subscribed to it (// on <asset>)
      → push a job for each
```

```
backend/windmill-queue/src/asset_dispatch.rs
```

Producer doesn't know its consumers. Consumers don't know each other.

Subscriptions alone define the graph.

How is this even safe to do?

Both halves need to agree on "what does this script read/write":

- **Backend**, at deploy → must be **authoritative** (it drives real job dispatch)
- **Frontend**, as you type → must be **instant** (live graph in the editor)
- **Testing a draft** → there are no asset rows yet, so the **frontend orchestrates the dev-run** — it's *authoritative* too, before anything is deployed

Two implementations = two sources of truth = drift = your **test** ≠ **prod**.

So we don't have two implementations.

We were already parsing every script

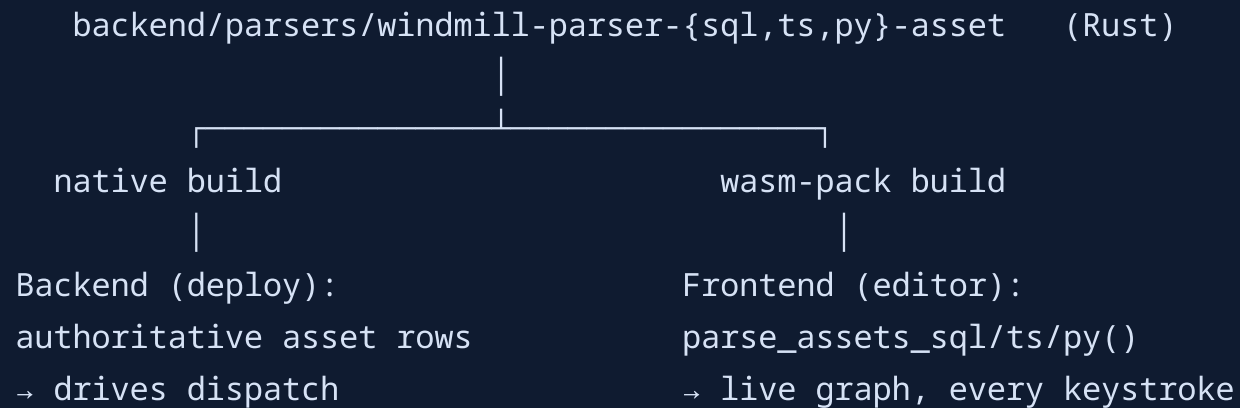
Every Windmill script is **already parsed** to derive its `main` signature — which powers the **input form** (params → JSON Schema → UI) and the **dependencies** (imports → lockfile). That parser already runs **native** (backend) + **WASM** (editor).

```
export async function main(  
  name: string,  
  count = 3,  
  db: Postgresql,           // a Windmill resource type  
) { /* ... */ }
```

↳ a JSON Schema → an auto-form: text field · number (default `3`) · resource picker.

The eureka: assets are just *one more visitor* on the same AST — no new infrastructure, just a parser we'd already shipped, hardened, and dual-compiled for years.

One parser, compiled twice



Same parser, one more visitor. SQL via a real DuckDB-dialect parser; TS/Python via real ASTs (track `ATTACH ... AS db / wmill.datatable('main')`, then resolve usages).

That's the whole trick

The WASM parser is what makes "no pipeline object" *viable*:

- It can **implement** the graph — backend dispatch off parsed assets
- It can **represent** the graph — frontend draws it live from the same parse

The script definitions *are* the pipeline.

The parser is the bridge that lets one artifact be both **executable** and **drawable**.

Demo: watch the graph build itself

1. Open a pipeline folder → **graph reconstructed from a SQL query**
2. Type `CREATE TABLE clean AS SELECT ... FROM orders`
→ WASM parses → **new node + edge appear, live**
3. Add `// on datatable://main/orders` → **subscription edge wires up**
4. **Deploy** → asset rows written → **run the source**
5. Watch the **cascade animate** down the graph (Activity feed groups the run)
6. Same DAG in the terminal: `wmill pipeline show f/orders`

The honest tradeoffs

Emergent graphs are great until they're spooky. So we made the implicit **visible**:

- **Cycle detection** — a write that loops back to its own producer is caught and broken, so cascades can't run away
- **AND-joins**: a script can wait for *all* inputs (partitions via `{partition}` token)
- **dispatch_event table** logs every decision: *dispatched / join-pending / skipped (why)* — surfaced on the job UI
- Backend re-parses at deploy → **server is authoritative**, browser can't lie
- Self-loops, debounce windows, retries — all declared in-script

Takeaways

Flows → explicit FSM. You write the graph; a generic interpreter walks it.

Pipelines → emergent graph. You write scripts; the graph *falls out* of them.

The unlock: **one parser, native + WASM**, so the script definitions are simultaneously what **runs** and what you **see**.

 **We're hiring** — Rust / TypeScript / systems engineers → windmill.dev/careers

Code: `windmill-queue/src/asset_dispatch.rs` · `parsers/windmill-parser-*-asset` ·
`frontend/.../AssetGraph` — all on `main`.

Appendix A — The real code

The actual source behind each claim. Excerpts are verbatim, lightly trimmed where noted.

A1 · "Subscriptions alone define the graph"

Straight from the dispatch module's own doc comment — **verbatim**:

```
//! When a script writes an asset and a downstream script subscribes to  
//! that asset via `// on s3://...`, this module pushes a job for each  
//! subscriber after the producer's job completes successfully. Any  
//! asset-writing top-level script cascades – there is no `// pipeline`  
//! gate on the producer side; subscriptions alone define the graph.
```

```
backend/windmill-queue/src/asset_dispatch.rs:11
```


A2 · The dispatch loop (lightly trimmed)

```
let producers = workspace_producer_writes(db, &job.workspace_id).await?; // cached
let Some(writes) = producers.get(runnable_path).cloned() else { return Ok(..) };

for (asset_kind, asset_path) in writes {
    let trigger_ref = format!("{}", prefix, asset_path);
    let subs = fetch_subscribers(db, &job.workspace_id, &trigger_ref).await?;
    for sub in subs {
        if sub.path == runnable_path { continue; } // skip self-loops
        if sub.join_all { /* AND-join: wait for all inputs */ }
        push_subscriber(db, job, &sub.path, asset_kind, &asset_path,
                        runnable_path, depth + 1, /* partition, debounce, retry */).await?;
    }
}
```

`asset_dispatch.rs::try_dispatch` (273). Non-producers cost **one in-memory lookup, zero queries**.

A3 · One function, native AND wasm

The **WASM export** is a four-line shim — it forwards to the same crate the backend links natively:

```
[cfg(feature = "asset-parser")]
#[wasm_bindgen]
pub fn parse_assets_sql(code: &str) -> String {
    match windmill_parser_sql_asset::parse_assets(code) { // ← same fn the backend calls
        Ok(r) => serde_json::to_string(&r).unwrap(),
        Err(e) => format!("err: {:?}", e),
    }
}
```

`windmill-parser-wasm/src/lib.rs` — same pattern for `parse_assets_ts` / `parse_assets_py`.

A4 · ...and it's a real parser

`parse_assets` isn't a regex — it runs the **actual DuckDB-dialect parser** and walks the AST:

```
pub fn parse_assets(input: &str) -> anyhow::Result<ParseAssetsOutput> {
    let statements = Parser::parse_sql(&DuckDbDialect, input)?;
    let mut collector = AssetCollector::new();
    for statement in statements {
        let _ = statement.visit(&mut collector);
    }
    // → reads/writes resolved from ATTACH bindings + table refs
}
```

`backend/parsers/windmill-parser-sql-asset/src/asset_parser.rs`

A5 · The hard case: data tables in TypeScript

A TS script reaches a data table through the SDK — the parser reads the **TypeScript** *and* the **SQL embedded inside it**:

```
import * as wmill from "windmill-client"
export async function main(since: string) {
  const sql = wmill.datatable("main")           // bind: sql → datatable "main"
  await sql`
    SELECT * FROM orders WHERE created_at > ${since}
  `.fetch()                                     // READ   main/orders
  await sql`
    CREATE TABLE daily_revenue AS
    SELECT day, sum(total) FROM orders GROUP BY day
  `.fetch()                                     // WRITE  main/daily_revenue
}
```

Inferred: `datatable://main/orders` **R** · `datatable://main/daily_revenue` **W** — the TS AST resolves the `sql` binding, lifts each template's SQL, and runs it through the **same SQL parser**. *A parser inside a parser.*

A6 · The browser side calls it on every keystroke

```
import initAssetParser, {
  parse_assets_ts, parse_assets_py, parse_assets_sql
} from 'windmill-parser-wasm-asset'

export async function inferAssets(language, code) {
  if (language === 'duckdb')      { await initWasmAsset(); return wrap(parse_assets_sql(code)) }
  if (language === 'bun' || 'deno'){ await initWasmAsset(); return wrap(parse_assets_ts(code)) }
  if (language === 'python3')     { await initWasmAsset(); return wrap(parse_assets_py(code)) }
}
```

`frontend/src/lib/infer.ts` — wired into `ScriptEditor.svelte` as a reactive `resource([lang, code])`.

Appendix B — Drawing the emergent graph

The graph has no stored layout either — it's computed every render.

B1 · Band-reserving tidy-tree layout

The graph is **reconstructed from a SQL query**, so it also has to be **laid out from scratch** each time. Naive layered (Sugiyama) layout let unrelated edges fight for columns. The fix, **verbatim from the source**:

```
// Each node reserves a horizontal band at least as wide as the sum of its
// children's bands ( $W(n) = \max(\text{NODE\_WIDTH}, \sum W(\text{children}) + \text{gaps})$ ) ...
// Bands are exclusive, so two nodes with different parents can never
// interleave horizontally – the failure mode of the previous Sugiyama layout.
//
// The band recursion stops at *join points*: a node with two or more parents
// belongs to no single parent's band ... Joins instead root their own subtree.
```

`frontend/.../AssetGraph/assetGraphLayout.ts` · falls back to a grid on cyclic input.

B2 · Edges route around obstacles

A long edge that skips layers would draw **through** unrelated node boxes. So we detour:

- sample the straight line at each intervening row
- if a node sits under it (within $\frac{1}{2} \cdot \text{width} + 8\text{px}$), it's a crossing
- route to the side the edge is already heading, outermost lane clears all crossings — **one detour, no box crossings**

`AssetGraphCanvas.svelte::detourForEdge` — computed **once per layout** (`$derived`), not per frame.

B3 · The live graph is a layered merge

What you see while editing = deployed graph + your unsaved reality, merged by precedence:

```
export function resolveGraph(input) {  
  const acc = seedAccumulator(input, ctx) // 1. base: /assets/graph (deployed)  
  seedDraftOverlays(acc, input) // 2. per-draft seeded triggers & outputs  
  applyLiveBufferOverlay(acc, input, ctx) // 3. the open script's live // on annotations  
  crossCheckSweptScripts(acc, input) // 4. reconcile renamed/removed  
  overlayInferredLineage(acc, input) // 5. WASM-inferred reads/writes (this keystroke)  
}
```

`resolveGraph.ts` — base < session-inferred < draft < **live annotations** (highest wins).